
Data Visualization GUI Interactivity Study

Booking (Ahead of time)

- Send consent form

Checklist (Preparation Before Study)

- A. 3 Copies Consent Form
- B. Charge any devices needed
- C. 2 Pens
- D. Verify with each of the X participants:
 - Time and place of meeting
 - Contact info:
 - Phone number
 - Address
 - Who will be there
 - Go over any consent questions if lingering
- E. Surveys:
 - 2 background questionnaires

Setup

- Set up design materials
 - Assign the participant an anonymous ID
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Consent (~10min)

Introduction:

Welcome! Thank you for participating in this study. Throughout the study, I may need to read from a script to ensure that I keep our session on track.

The purpose of this study is to investigate user interactions with dashboard widgets, graphs, the carousel, sidebars, and the overall data visualization pipeline in a web-based AI-driven platform. By studying how participants engage with these features, we aim to improve their design and usability.

This study is being conducted by myself, George Morales.

Important Information

Many of the study tasks involve answering questions about your background and experience with data visualization tools. You will also be asked to interact with various dashboard elements, such as uploading datasets, labeling columns, generating graphs, customizing widgets, and navigating the carousel. Additionally, we will conduct a semi-structured interview to discuss your experience, challenges, and preferences.

Your participation is entirely voluntary, and you may choose to withdraw at any time. If you have any questions before we begin, feel free to ask.

Study Procedure

Today's session consists of two main activities.

First, I will ask you to upload a dataset, describe it, and label its columns for visualization.

Next, I will introduce you to the dashboard workspace, where you can interact with data visualizations, graphs, and charts. You will have the opportunity to explore the interface, adjust settings, and experiment with different configurations to see how the system responds.

Then, I will ask you to create and customize dashboard widgets, rearrange elements within the carousel, and interact with the sidebar to refine the display of your visualizations.

Finally, I will ask you to provide feedback on your experience, share any challenges or preferences you encountered, and discuss potential improvements to the interaction design.

Design Presentation (~5-10min.)

This will be done on my laptop. The browser-based GUI design consists of two main parts: data upload and annotation and dashboard interactions. Users start by uploading a dataset (CSV/Excel), labeling columns, and adding descriptions for context. Once processed, they enter the interactive dashboard, where they can generate and customize graphs, explore widgets, and navigate visualizations within a carousel. Sidebars provide additional options for filtering and refining insights.

Interview (~15min.)

I'd like to ask you a few questions about your experience interacting with the dashboard, including the data upload process, annotation, and visualization features. Please share your thoughts as openly as possible.

- 1. *How intuitive was the data upload and annotation process? Were there any challenges or areas for improvement?***

Uploading the dataset was quick, and the annotation process was fairly straightforward. The system provided clear prompts, which made it easy to understand what needed to be done. However, for datasets with many columns, the process became repetitive. A more dynamic approach, like autosuggestions for labels could be beneficial.

- 2. *How easy was it to navigate and interact with the dashboard, widgets, graphs, and carousel?***

Overall, the interface was easy to navigate, and interacting with the widgets and graphs felt intuitive. However, the rigid layout made it feel restrictive. The carousel was helpful for browsing visualizations, but it would be better if it could be minimized when not in use. Allowing users to rearrange widgets freely instead of keeping them locked in place would improve customization and workflow.

- 3. *Were the customization options sufficient for refining visualizations? If not, what additional features would you find useful?***

The customization options were decent, and features like the info tab, widget expansion, and the ability to pin and unpin widgets to move them back to the carousel were especially useful. These made it easy to manage visualizations without feeling cluttered. However, more flexibility in adjusting widget sizes and positions would improve usability.

- 4. *Did anything about the interface feel confusing, frustrating, or particularly helpful? Why?***

The interface felt well-structured, and the workflow was easy to follow. The pinning and unpinning features made managing widgets seamless, and the expansion option was useful for focusing on specific visualizations. The carousel provided a smooth way to organize and access different graphs.

Conclusion

Thank you for your help with our study. We are finished with all the session's tasks.